

MST-Technologies and Processes

March, 25th, 2022

Duration: 120 min.

Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task
- write with a pen NOT with a pencil
- if you hinder the exam you will be removed from the lecture hall and fail the exam (5.0 with the statement "Störung")

Cheating

- any attempt of cheating results in immediate failure of the exam (5.0 with the statement "Täuschung")
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
 - in case you feel unwell during the exam, you may withdraw during the exam for medical reasons; you need to provide (within 3 days) a written note from your doctor confirming the withdrawal for medical reasons
 - please see:
<https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams>
 - you find the relevant forms here:
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>
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Necessary Points: **50 of 100 in total**

Points: _____

Result: _____

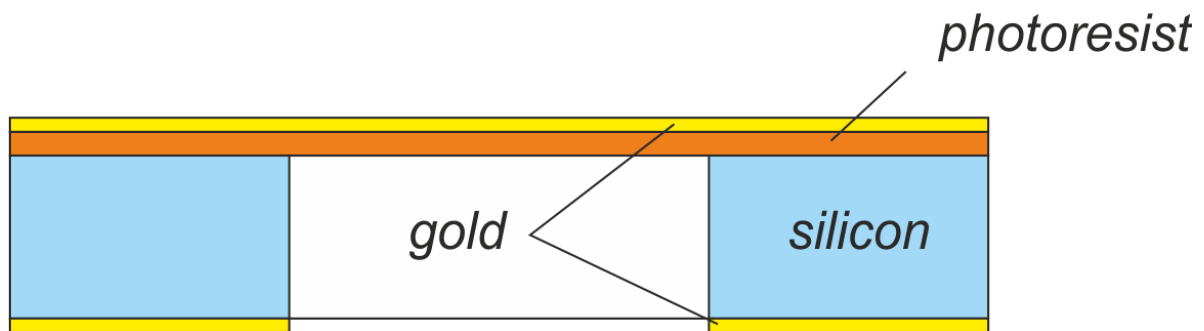


1. Processes

25

Sketch the manufacturing steps required to fabricate the following structure. The device is a polymeric actuator which has a suspended layer of an elastic polymer (here: a photoresist) sandwiched between two electrode layers made from gold. The basis substrate is a silicon wafer.

Sketch and briefly explain each step. Each manufacturing process (e.g., lithography) used should be depicted out entirely when you first use it. If you use the same process again, you do not need to redraw the entire process.



2. Cleanroom Technology

25

- 2.1 We usually use ISO norms to classify the quality of a cleanroom. Which is the best cleanroom quality level - ISO 1 or ISO 6? 2
- 2.2 Name four typical environmental parameters, which are controlled in a cleanroom. 4
- 2.3 The following image shows a very common type of cleanroom setup:



- Which type of cleanroom is this? 2
- Name two distinct advantages of this setup. 2
- 2.4 Name five typical recommendations for working in a cleanroom (cleanroom etiquette). 5
- 2.5 Filter-fan units (FFUs) make use of four distinct filtering effects. Sketch and briefly explain how each of these effects works. 10

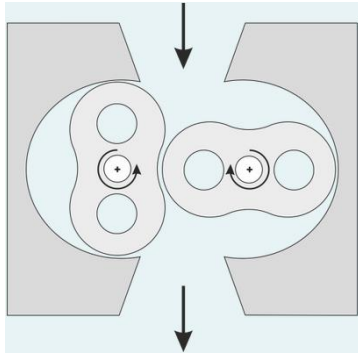


3. Vacuum technology

18

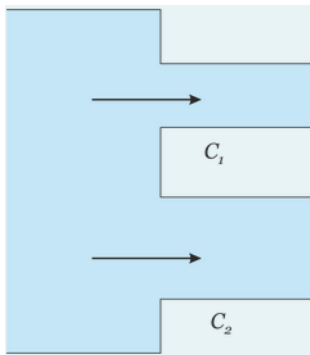
3.1 Which type of vacuum pump is depicted here?

2



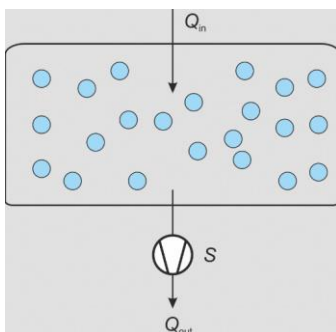
3.2 A hydraulic element with the conductance $C_1 = 7.8 \frac{\text{m}^3}{\text{s}}$ is connected in parallel to a second hydraulic element with the conductance $C_2 = 6.5 \frac{\text{m}^3}{\text{s}}$. What is the overall conductance C_{tot} of the system?

4



3.3 A gas leak adds a standardized (at a given pressure) flow rate of $8.4 \frac{\text{cm}^3}{\text{s}}$ mbar to an evacuated chamber kept at $p = 5$ mbar. What is the pumping speed S required to sustain this pressure?

4



3.4 Sketch and briefly explain the turbomolecular pump.

5

Why does the turbomolecular pump always require a backing pump?

3



4. Thermal oxidation of silicon

12

4.1 In the Deal-Grove model, the following variables are used:

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- D as the diffusivity of the oxidant in silicon dioxide
- k as the reaction rate of the oxidation of silicon to silicon dioxide
- N as the number of oxidant molecules required to produce one unit volume of oxide
- C as the ambient concentration of the oxidant
- $A = \frac{2D}{k}$ as a simplification factor
- $B = \frac{2DC}{N}$ as a second simplification factor

State the equation for the layer thickness $x(t)$ at higher layer thicknesses, i.e., for $x(t > t_c)$.

4.2 For usage in the Deal-Grove model we determined the following factors:

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- $A = 2 \mu\text{m}$
- $B = 0.25 \mu\text{m}^2/\text{h}$

What is the characteristic time t_c ? [You do not need a calculator for this - if you find odd numbers, just write down the equation with the numbers]

4.3 Using the same parameters as above, we now run an oxidation process for 4 hours (which is longer than t_c). What is the expected layer thickness we achieve? [You do not need a calculator for this - if you find odd numbers, just write down the equation with the numbers]

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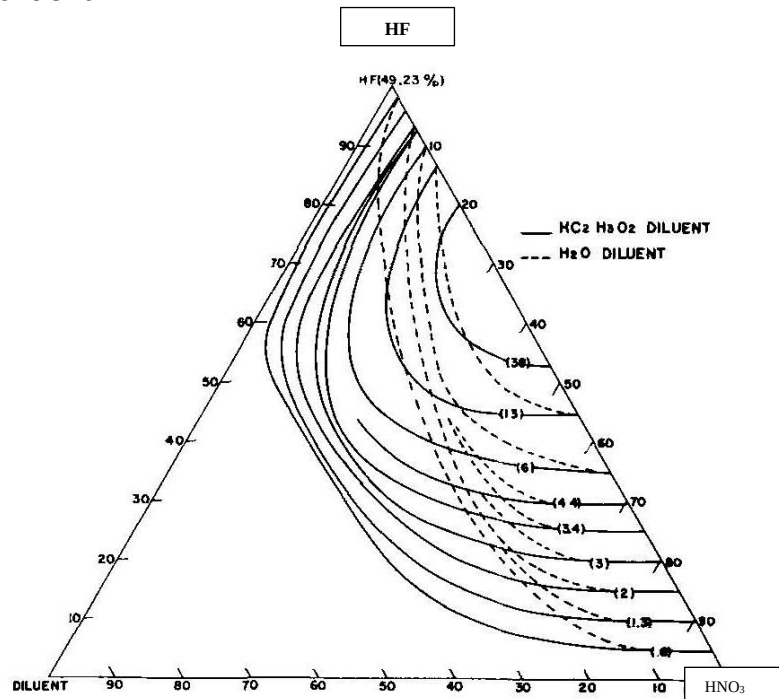


5. Etching

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- 5.1 Given the following etching diagram state three etching solutions which give an etching rate of 6 ($\mu\text{m}/\text{min}$) in acetic acid (solid lines) as diluent:

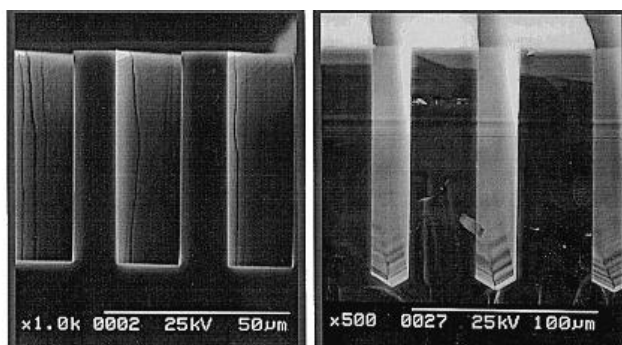
6



- 5.2 Name two typical etching alkaline etchants for anisotropic silicon etching.
- 5.3 The following is an image of a (110) wafer having being etched by an alkaline etchant. The right image shows the (111) crystal planes at the bottom of the trench, the left one does not. Which etchant has been used in the left image?

2

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- 5.4 Sketch and describe the process steps of the Bosch process. Take care to also write down the actual etching reaction of silicon.

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Laboratory of Process Technology

Name: _____
Matriculation number: _____

